

Rohanraje Bhosale

AI/ML - Data Scientist

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Summary

Data Scientist and Machine Learning Engineer with experience building predictive models, time-series forecasts, and production-grade ML systems across structured and unstructured data. Strong in Python, SQL, PyTorch, and scalable ML pipelines covering data ingestion, feature engineering, model training, evaluation, deployment, and monitoring. Proven ability to design experiments, debug model failures, and translate analytical insights into business and product decisions across media, healthcare, security, and GenAI applications.

Core Technical Skills

Programming & ML:	Python, SQL, PyTorch, NumPy, pandas, scikit-learn, PySpark
Modeling & Analytics:	Regression, Classification, Time-Series Forecasting, Anomaly Detection, A/B Testing
GenAI & Agents:	LLMs, RAG Systems, Agentic AI, Embeddings, Vector Databases
Data System:	Spark, PostresSQL, MongoDB, BigQuery, Parquet,
ML Systems:	FastAPI, Streamlit, Docker,

Experience

BOTLOT - Data Science Intern

May 2025 – August 2025

Worked on production ML and analytics systems with emphasis on reliability and performance; analyzed large-scale behavioral datasets to identify engagement trends and anomalies; built SQL pipelines for recurring reporting and ad-hoc analysis; developed predictive models for user behavior and system performance; supported A/B testing, KPI definition, and feature impact analysis; deployed FastAPI-based inference services and collaborated with engineers to integrate ML into production workflows.

Project 550 - University of Massachusetts Dartmouth – Machine Learning Engineer

January 2025 – December 2025

Extracted, cleaned, and validated 6.4GB+ multi-source datasets for viewership and engagement forecasting; performed EDA, variance analysis, and hypothesis testing; built regression, time-series, and ML models for planning and optimization; designed forecast accuracy metrics, drift detection, and dashboards communicating confidence intervals and business impact.

University of Massachusetts Dartmouth - Student Data Operations Analyst

January 2024 – December 2025

Automated Python-based operational analytics across 200+ daily transactions; maintained 99%+ data accuracy; identified operational bottlenecks and delivered insights improving turnaround time and service efficiency.

Sahyadri Polytechnic Sawarde - Assistant Professor (Data Science)

June 2023 – January 2024

Designed Python- and SQL-based analytics systems, applied regression, classification, and statistical analysis to academic datasets. Built batch-processing systems for grading and plagiarism detection; mentored 12+ technical teams and enforced data quality and reproducibility practices.

Trismus Healthcare Services - Machine Learning Engineer (Medical Imaging)

January 2022 – June 2023

Built preprocessing and feature engineering pipelines for medical imaging datasets; developed and evaluated CNN-based diagnostic models; performed data audits and collaborated with cross-functional teams to prepare models for deployment in regulated environments.

Projects (GitHub)

ROHbot (RAG Portfolio Assistant) - Built a production-grade retrieval-augmented generation system combining LLMs with a vector database to answer portfolio and project queries; implemented embedding-based retrieval, prompt orchestration, and API-driven inference for reliable, low-latency responses ([link](#))

Agentic AI System - Designed an agent-based AI system orchestrating multi-step reasoning and tool use; implemented LLM-driven workflows, task planning, and structured outputs for automated analysis and decision support. ([link](#))

Network Anomaly Detection System - Built ML-based detection pipeline using Isolation Forest and supervised models; analyzed malicious traffic patterns; developed dashboards for anomaly scores and threat visibility. ([link](#))

AI Data Analyst System - Developed an LLM-powered analytics assistant to query, summarize, and explain datasets; integrated Python-based analysis workflows for automated insight generation. ([link](#))

Car Price Prediction Model - Built regression models for price estimation using structured vehicle data; performed feature engineering, model evaluation, and error analysis. ([link](#))

Data Quality & Validation System - Implemented automated data validation checks, consistency rules, and anomaly flags to ensure dataset reliability for downstream analytics and ML pipelines. ([link](#))

Education, Research & Certifications

University of Massachusetts Dartmouth — M.S. Data Science

December 2025

Publication: Bibliometric Analysis of Machine Learning and Text Mining Algorithms for Diagnosis of Leukemia.

Certifications: Modern Computer Vision; PyTorch; TensorFlow2/Keras & OpenCV; PMI-ACP (Agile).